

01 / PURPOSE AND OVERVIEW

Multilingual news recommendation

Write an interest, discover relevant historical news, then inspect the facts and meaning signals behind the recommendation.

ONE ARTICLE, TWO PATHS



DEVELOPMENT SEQUENCE / AUGUST TO OCTOBER 2026

01

Prototype

Expose each model stage

02

Real data

Collect, label and review

03

Validation

Choose the extraction adapter

04

Release

Index 300 articles and host

[Try the live demo](#)[Read the code](#)

02 / DATA PREPARATION

Real articles and traceable labels

Source attribution and split discipline make the training and evaluation interpretable. Collection, training and release sizes serve different purposes.

COLLECTION

5,000 historical Wikinews articles

3,539 English / 1,461 Arabic

**PREPARATION**

Clean and group before splitting

Remove boilerplate and duplicates; group related versions. Retain source URL, revision, contributor attribution, license and content hash.

**TRAINING INPUT**

30 accepted examples

Machine-assisted article-to-JSON labels accepted from 160 attempts.

HUMAN EVALUATION

60 completed review items

30 extraction references plus 30 recommendation judgments.

Published hyperlinks provide incomplete mention annotations. They cannot certify that an article has no other entities. Machine-assisted empty lists remain weak judgments.

The archive uses 2019 snapshots referenced by Mewsli-9. This custom reconstruction is not the official Mewsli benchmark. Translation grouping is heuristic; residual overlap is possible.

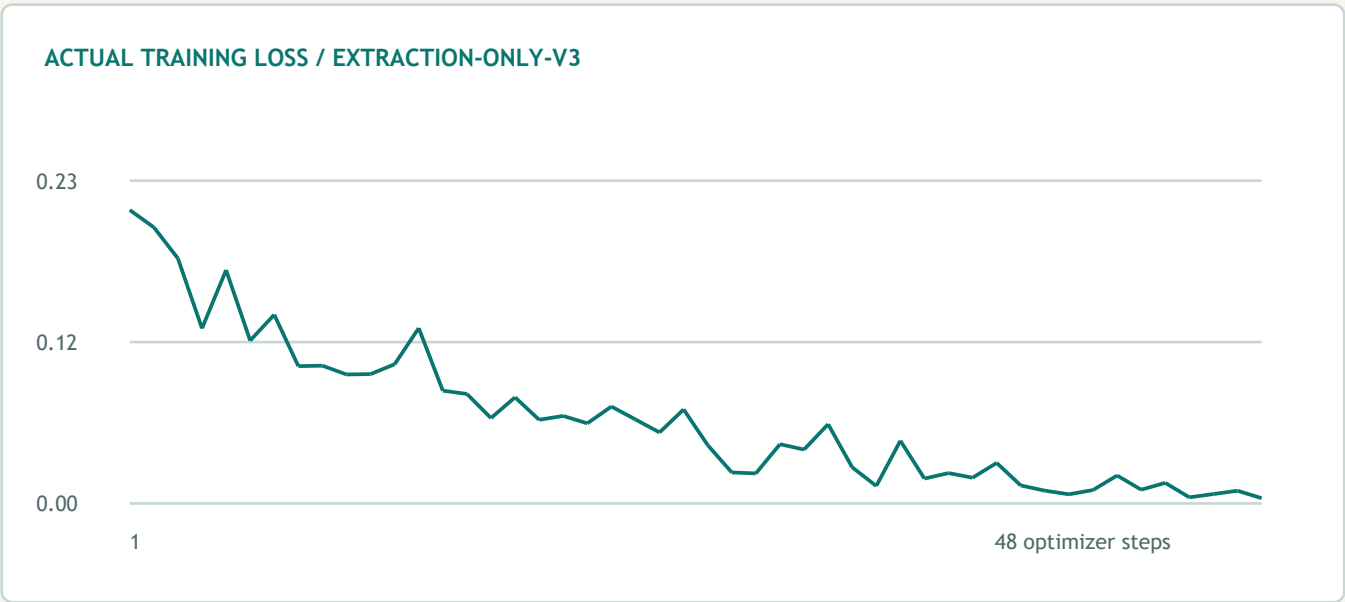
[Source and license details](#)

[Collection code](#)

Teach Qwen to produce structured facts

Qwen2.5-3B supplies language knowledge. QLoRA freezes the base in 4-bit memory and trains small LoRA updates using article/answer examples. The tokenizer vocabulary stays unchanged.

Text > token IDs > Qwen + trained adapter > extraction JSON



30 examples seen; 288 adapter tensors changed. Loss is the mean of four supervised-token microbatch means per update. Different batches vary; lower training loss alone does not establish accuracy.

10 validation articles	Entity F1	Schema valid
Base Qwen	0.289	40%
Extraction only (selected)	0.372	90%
Domain + extraction	0.354	90%

Selection required schema validity $\geq \max(80\%, \text{base})$, improved overall F1 and no language F1 regression. Domain adaptation was evaluated but not selected. Earlier interrupted and unsuccessful runs remain in the evidence history.

[Training notebook](#)

[Shared trainer](#)

Connect mentions using context

A generated name is not an entity ID. E5 retrieves candidates from a limited Wikidata catalogue; a supervised logistic classifier ranks them using contextual and name features.

ACTUAL RECORDED ARTICLE / EN-39534

Article mentions "U.S. President George W. Bush" during a report about Iran nuclear diplomacy. Qwen extracted "George W. Bush" as a person.



Retrieved candidate	Learned score
George W. Bush	0.9851
George Bush	0.6088
George Speight	0.0000

ACCEPTED

George W. Bush / Q207

The top candidate passes the acceptance rule.

UNRESOLVED

Donald Rumsfeld

Low candidate scores leave this mention without an ID.

On the previously reported article-disjoint holdout, reused for this release (200 mentions): 95.9% precision among accepted links, with 36.5% coverage (73 accepted). Abstention reduces false links but limits coverage.

Scores are not calibrated probabilities. The catalogue covers a subset of Wikidata; unseen and ambiguous aliases remain difficult. Source text supports a mention, but does not prove every generated relationship.

05 / RECOMMENDATION

Compare meaning, then add constraints

E5 compares the user's written interest with natural article text. It uses pretrained multilingual knowledge directly, so Arabic and English can share a meaning space without manual translation.

QUERY INPUT

Iran nuclear diplomacy and international sanctions

ARTICLE INPUT

Historical article text

Title and body use the passage: prefix. Article vectors are computed once.

INTEREST INPUT

Written interest

The interest uses the query: prefix. A new interest gets a fresh vector.



768 values per normalized vector

Dot product = cosine similarity. E5 receives natural text, independently of the extraction JSON.

Without filters, rank by E5 similarity. With filters, complete matches rank first; within each group, sort by 75% E5 similarity + 25% exact filter coverage. Both signals remain visible in the app.

Recorded first result: Peace activist claims Iran keen to compromise on nuclear issue, Cheney, Rumsfeld allegedly block negotiations. Similarity/ranking score 0.836 for this interest. The number orders results; it is not a confidence percentage.

Dynamic phrase probes score smaller text pairs to illustrate related passages. They do not identify the internal reason for a whole-article score.

[Ranking and shared pipeline](#)

[Recommendation notebook](#)

06 / RESULTS AND REPRODUCTION

Measured performance and next steps

The release works end to end, with limited model quality. Frozen references and saved predictions make the reported measurements reproducible.

EXTRACTION / 20 HUMAN-REVIEWED TEST ARTICLES

Parsing valid	Schema valid	Precision	Recall	Entity F1
100%	95%	49.0%	23.0%	0.313

Ten English and ten Arabic articles, held out from our training. Entity F1 uses normalized surface names; topics and relationships are schema-checked but not human-scored.

RETRIEVAL / TWO TEST QUERIES, TEN JUDGED ARTICLES EACH

Method	nDCG@5	Recall@5
Keyword	0.612	0.500
E5	0.956	0.500
Hybrid	0.956	0.500

Keyword and E5 rank all ten candidates per pool; hybrid ranks only eight and nine after extraction failures. These tiny pools do not establish full-corpus superiority. E5 and hybrid have equal reported averages.

A FAILURE TO INSPECT

The hosted model classified Beirut as a country. A European energy/gas query also returned an unrelated Google-slander article near the top. Failures remain visible alongside successful examples.

Frozen quality metrics use local 4-bit Qwen; hosted inference uses the same adapter with float16 base weights. Quality parity has not been measured. Free GPU quotas may interrupt inference.

[Reproduce saved metrics and open notebooks](#)

[Results and limitations](#)

[Data, licenses and model versions](#)